

NetSage AI — the interface

Every screen of the running application, captured end to end

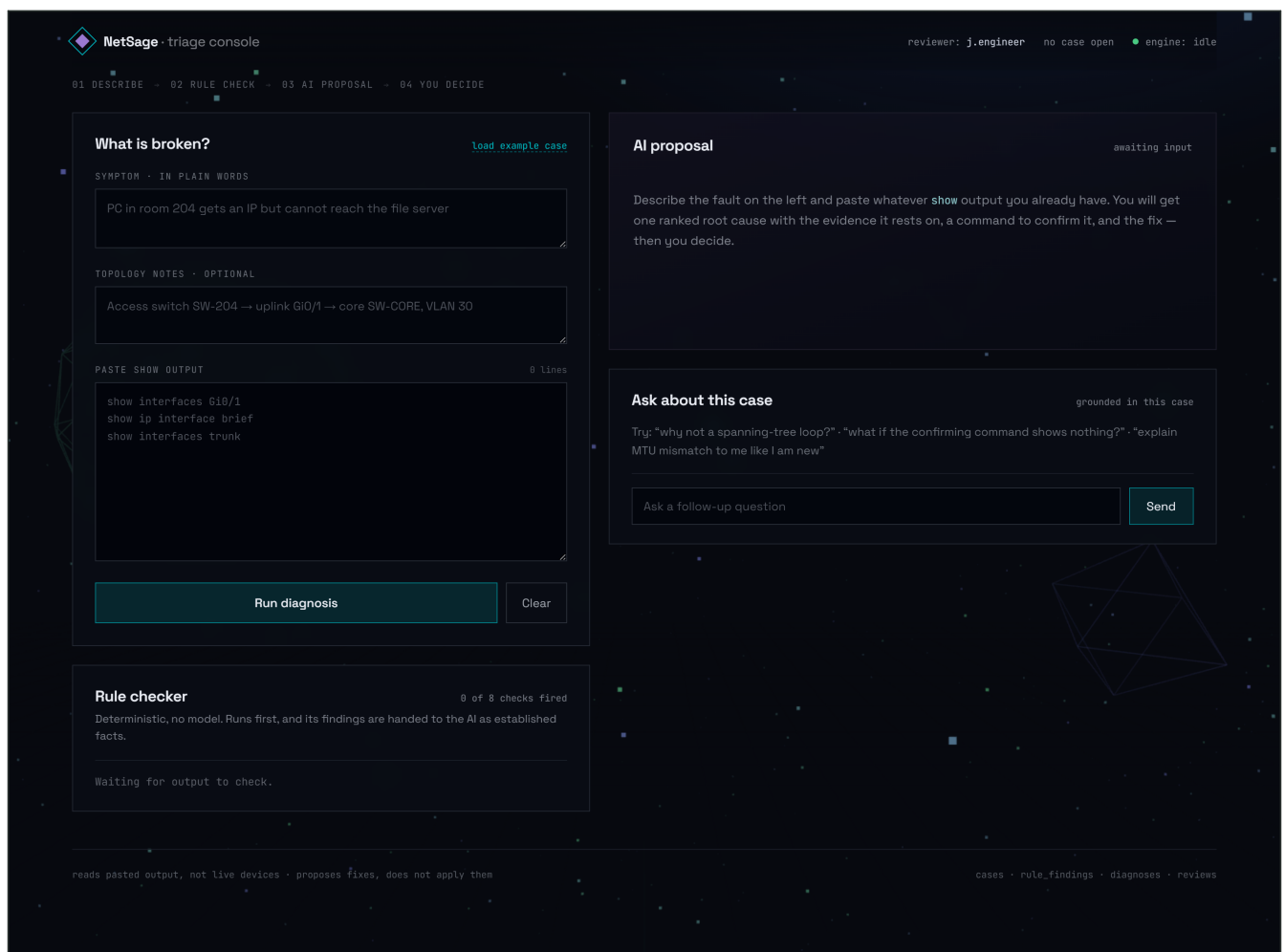
Submitted by	Danish Soreng
Institution	KIIT University
Captured from	The running Flask application at 127.0.0.1:5000, via headless Chrome
Engine shown	heuristic — the offline fallback, so no API key is needed to reproduce this

The eight captures that follow are screenshots of the real application, not mock-ups. Pages 2 to 5 follow a single fault through the whole workflow: an empty console, a case entered, the deterministic findings and the AI proposal that was built on top of them, and finally the human decision that closes the case. Pages 6 to 9 show the remaining screens.

Each screenshot is the full page, not a cropped viewport.

1. The triage console

The screen a reviewer opens. Three inputs on the left - the symptom in plain words, optional topology notes, and whatever command output has been collected. The strip along the top names the four stages the case will pass through, and the footer states permanently that the system reads pasted output rather than live devices.



2. A case entered

The example case loaded. PC1 sits on 192.168.10.25 with mask 255.255.255.0, so its network is 192.168.10.x - but its default gateway is 192.168.20.1, on a different network. PC2 beside it is identical apart from a correct gateway. Nothing has been submitted yet; the header still reads “no case open”.

The screenshot displays the NetSage triage console interface. At the top, the header shows "NetSage · triage console" and a status bar with "reviewer: j.engineer", "no case open", and "engine: idle". Below the header, a progress bar indicates the current step: "01 DESCRIBE" (active), followed by "02 RULE CHECK", "03 AI PROPOSAL", and "04 YOU DECIDE".

The main interface is divided into several sections:

- What is broken?**: This section contains a "SYNOPSIS · IN PLAIN WORDS" box with the text: "PC1 in Lab 2 cannot open the file server. PC2, plugged into the next port on the same switch, works perfectly." Below this is a "TOPOLOGY NOTES · OPTIONAL" box with network configuration details for PC1, PC2, and FS1. A "PASTE · SHOW OUTPUT" box displays the output of commands run on PC1, including ipconfig and ping results. A "Run diagnosis" button is located at the bottom of this section.
- AI proposal**: This section is titled "AI proposal" and has a status of "awaiting input". It contains a prompt: "Describe the fault on the left and paste whatever show output you already have. You will get one ranked root cause with the evidence it rests on, a command to confirm it, and the fix — then you decide."
- Ask about this case**: This section is titled "Ask about this case" and has a status of "grounded in this case". It contains a prompt: "Try: 'why not a spanning-tree loop?'. 'what if the confirming command shows nothing?'. 'explain MTU mismatch to me like I am new'". Below the prompt is a text input field labeled "Ask a follow-up question" and a "Send" button.
- Rule checker**: This section is titled "Rule checker" and has a status of "0 of 8 checks fired". It contains a message: "Deterministic, no model. Runs first, and its findings are handed to the AI as established facts." Below this is a message: "Waiting for output to check."

At the bottom of the interface, there is a footer with the text: "reads pasted output, not live devices · proposes fixes, does not apply them" and a navigation bar with links: "cases · rule_findings · diagnoses · reviews".

3. Rule findings and the AI proposal

/api/triage

One press of Run diagnosis. The deterministic checker ran first and raised GATEWAY_OFF_SUBNET as a high-severity fact, quoting the exact line it fired on. That finding was then passed to the AI as established truth, and the proposal on the right names the root cause with its layer, its confidence, the evidence it rests on, a command to confirm it, and the fix. The header now reads “case LIVE-0001 saved” - the case is in the database.

NetSage · triage console

reviewer: j.engineer case LIVE-0001 saved engine: heuristic

01 DESCRIBE 02 RULE CHECK 03 AI PROPOSAL 04 YOU DECIDE

What is broken?

load example case

SYNOPSIS · IN PLAIN WORDS

PC1 in Lab 2 cannot open the file server. PC2, plugged into the next port on the same switch, works perfectly.

TOPOLOGY NOTES · OPTIONAL

PC1 192.168.10.25 255.255.255.0 192.168.20.1 vlan10
PC2 192.168.10.26 255.255.255.0 192.168.10.1 vlan10
FS1 192.168.30.10 255.255.255.0 192.168.30.1 vlan30

PASTE SHOW OUTPUT 15 Lines

PC1> ipconfig
IPv4 Address : 192.168.10.25
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.20.1

PC1> ping 192.168.10.26
Reply from 192.168.10.26: bytes=32 time<1ms TTL=128

PC1> ping 192.168.30.10
Reply from 192.168.10.25: Destination host unreachable

Run diagnosis again Clear

Rule checker

1 of 8 checks fired

Deterministic, no model. Runs first, and its findings are handed to the AI as established facts.

GATEWAY_OFF_SUBNET HIGH
PC1's default gateway 192.168.20.1 is outside its own subnet 192.168.10.0/24.
PC1 192.168.10.25/255.255.255.0 → gateway 192.168.20.1

AI proposal

engine: heuristic

routing L3 confidence 0.90

The default gateway is not inside the host's own subnet, so nothing can leave the subnet.

EVIDENCE

PC1 192.168.10.25/255.255.255.0 → gateway 192.168.20.1

RUN THIS TO CONFIRM

show run interface COPY

THEN FIX IT

01 Correct the item named above, then re-test connectivity end to end.

Your decision

Nothing is recorded until you choose. Editing keeps the AI's version alongside yours in the log.

Accept Edit Reject

REVIEWER NOTES

What you actually found on the device

Ask about this case

grounded in this case

Try: "why not a spanning-tree loop?" - "what if the confirming command shows nothing?" - "explain MTU mismatch to me like I am new"

Ask a follow-up question Send

reads posted output, not live devices · proposes fixes, does not apply them cases · rule_findings · diagnoses · reviews

4. The human decision

/api/review

Accept has been pressed and the review log at the foot of the page records it, timestamped. This is the only action in the entire system that closes a case: until a person presses one of these three buttons the proposal stays marked pending, and no code path anywhere marks it correct on its own.

NetSage · triage console

reviewer: j.engineer case LIVE-0001 saved engine: heuristic

01 DESCRIBE 02 RULE CHECK 03 AI PROPOSAL 04 YOU DECIDE

What is broken?

load example case

SYMPTOM · IN PLAIN WORDS

PC1 in Lab 2 cannot open the file server. PC2, plugged into the next port on the same switch, works perfectly.

TOPOLOGY NOTES · OPTIONAL

PC1 192.168.10.25 255.255.255.0 192.168.20.1 vlan10
PC2 192.168.10.26 255.255.255.0 192.168.10.1 vlan10
E1 102.168.30.1 255.255.255.0 102.168.30.1 vlan30

PASTE SHOW OUTPUT 15 lines

PC1> ipconfig
 IPv4 Address : 192.168.10.25
 Subnet Mask : 255.255.255.0
 Default Gateway : 192.168.20.1

PC1> ping 192.168.10.26
Reply from 192.168.10.26: bytes=32 time<1ms TTL=128

PC1> ping 192.168.30.10
Reply from 192.168.10.25: Destination host unreachable

Run diagnosis again Clear

Rule checker

1 of 8 checks fired

Deterministic, no model. Runs first, and its findings are handed to the AI as established facts.

GATEWAY_OFF_SUBNET HIGH
PC1's default gateway 192.168.20.1 is outside its own subnet 192.168.10.0/24.
PC1 192.168.10.25/255.255.255.0 → gateway 192.168.20.1

AI proposal

engine: heuristic

routing L3 confidence 0.98

The default gateway is not inside the host's own subnet, so nothing can leave the subnet.

EVIDENCE

PC1 192.168.10.25/255.255.255.0 → gateway 192.168.20.1

RUN THIS TO CONFIRM

show run interface COPY

THEN FIX IT

01 Correct the item named above, then re-test connectivity end to end.

Your decision

Nothing is recorded until you choose. Editing keeps the AI's version alongside yours in the log.

Accept Edit Reject

REVIEWER NOTES

What you actually found on the device

Ask about this case

grounded in this case

Try: "why not a spanning-tree loop?" - "what if the confirming command shows nothing?" - "explain MTU mismatch to me like I am new"

Ask a follow-up question Send

Review log

23:31 ACCEPTED The default gateway is not inside the host's own subnet, so nothing can leave the subnet.

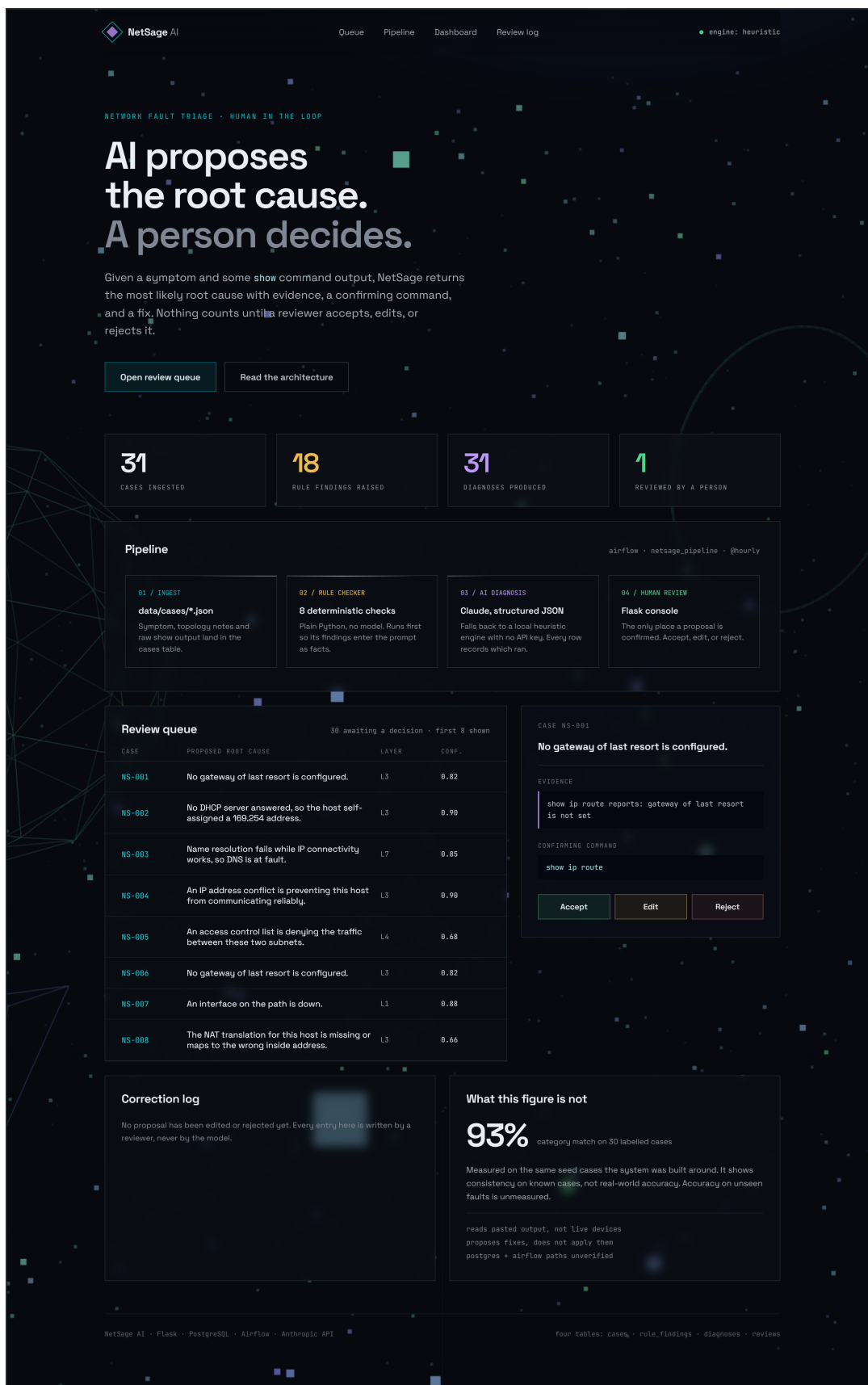
reads posted output, not live devices · proposes fixes, does not apply them

cases · rule_findings · diagnoses · reviews

5. Pipeline overview

/overview

The wide view, with every figure read live from the database on load. The four counters correspond to the four tables; below them the pipeline stages, the queue of cases still awaiting a decision, a spotlight case ready to be judged, the correction log, and the panel that states plainly what the agreement figure is not.



6. Review queue

/queue

Every case with its current review state, tallied across the top and colour-coded by OSI layer. This is the working list a reviewer opens at the start of a shift.

NetSage AI / review console

ConsoleQueueDashboard

Review queue

Every case below has an AI proposal waiting on a human decision. Nothing here counts as a confirmed fault until someone reviews it.

No API key is set, so diagnoses were produced by the local heuristic engine. Set ANTHROPIC_API_KEY and re-run the pipeline to use the model.

30	1	0	0
PENDING	ACCEPTED	EDITED	REJECTED

LIVE-0001	PC1 in Lab 2 cannot open the file server. PC2, plugged into the next port on the same switch, works ...	LAYER 3	accepted
NS-001	PC1 received an IP address but cannot reach the file server FS1 on another VLAN. Ping to its own de...	LAYER 3	pending
NS-002	A new employee's PC shows limited connectivity and an address starting 169.254.	LAYER 3	pending
NS-003	Users report the intranet portal is down, but it works when they type the IP address directly.	LAYER 7	pending
NS-004	Two desks in accounts intermittently lose network access; both work fine when the other is unplugg...	LAYER 3	pending
NS-005	Guest Wi-Fi users can reach the internal HR file share, which should not be possible.	LAYER 4	pending
NS-006	A branch office PC can reach other machines in the branch but nothing at head office.	LAYER 3	pending
NS-007	A newly cabled desk gets no link light and no IP address at all.	LAYER 1	pending
NS-008	A vendor cannot reach the newly published web server from outside the office.	LAYER 3	pending
NS-009	After a desk move a PC gets an IP in the wrong range and cannot reach its department server.	LAYER 2	pending
NS-010	A laptop brought back from home cannot reach anything on the office network.	LAYER 3	pending
NS-011	Half of a relocated team got IP addresses, the other half did not.	LAYER 3	pending
NS-012	Devices on VLAN 20 can ping each other but cannot reach any other VLAN.	LAYER 2	pending
NS-013	A workstation cannot reach anything outside its own subnet after a manual IP change.	LAYER 3	pending
NS-014	Printing to the network printer fails from one department only.	LAYER 4	pending
NS-015	A server was re-addressed and now nobody can reach it by name.	LAYER 7	pending
NS-016	Internet browsing fails for all users while internal resources work normally.	LAYER 3	pending
NS-017	Two hosts that should be in the same VLAN are on different subnets and cannot see each other.	LAYER 2	pending
NS-018	VoIP handsets never come online after a switch replacement.	LAYER 2	pending
NS-019	One PC can browse the web but cannot reach the internal ERP server.	LAYER 4	pending
NS-020	Outbound internet works but a remote support tool cannot connect back in.	LAYER 3	pending
NS-021	Wireless clients get addresses but wired clients on the same switch do not.	LAYER 3	pending
NS-022	A link between two switches came up but no traffic passes between buildings.	LAYER 2	pending
NS-023	Users report the network is slow and one uplink shows constant errors.	LAYER 3	pending - low conf
NS-024	A test machine cannot reach the internet, though its neighbour on the same desk can.	LAYER 3	pending
NS-025	After a firewall change, file transfers to the backup server hang partway through.	LAYER 4	pending
NS-026	New subnet was added but its hosts cannot reach any other subnet.	LAYER 2	pending
NS-027	Some websites load and others do not, with no obvious pattern.	LAYER 7	pending
NS-028	A camera system on its own VLAN lost connectivity to the recorder after a config change.	LAYER 1	pending
NS-029	Two departments were merged and now several machines report address conflicts.	LAYER 3	pending
NS-030	An external partner can reach the web server but not the API server behind the same public address.	LAYER 3	pending

7. Case detail and review form

/case/NS-004

One case read top to bottom: the evidence collected, the deterministic checks that fired, the full AI proposal, and the review form. Case NS-004 is an address conflict - two hosts configured with 172.16.4.50 - which the checker identifies with certainty rather than as an opinion. For the evaluation set the recorded true cause is shown at the foot for comparison.

NetSage AI / review console

ConsoleQueueDashboard

CASE NS-004 · SEVERITY HIGH

Two desks in accounts intermittently lose network access; both work fine when the other is unplugged.

[← Back to the queue](#)

Evidence collected

TOPOLOGY NOTES

PC11 172.16.4.50 255.255.255.0 172.16.4.1 v1an40
PC12 172.16.4.50 255.255.255.0 172.16.4.1 v1an40

COMMAND OUTPUT

SW2# show ip arp
Internet 172.16.4.50 0 00E0.1234.AAAA ARPA
Internet 172.16.4.50 0 00E0.5678.BBBB ARPA

Deterministic checks

Plain Python, no model involved. These are mechanical facts about the configuration, so they cannot be hallucinated.

HIGH DUPLICATE_IP
IP address 172.16.4.50 is assigned to more than one device.
Devices sharing 172.16.4.50: PC11, PC12

AI proposal - engine: heuristic

ROOT CAUSEAn IP address conflict is preventing this host from communicating reliably.

FAULT TYPEOTHER

OSI LAYER

LAYER 3

CONFIDENCE

90%

EVIDENCE USEDDevices sharing 172.16.4.50: PC11, PC12

NEXT COMMANDshow ip interface brief

SUGGESTED FIXCorrect the item named above, then re-test connectivity end to end.

Human decision

This proposal has not been reviewed. It is not a confirmed fault yet.

DECISION

Accepted

Edited

Rejected

REVIEWER

Your name

CORRECTED ROOT CAUSE (fill in if you edited or rejected)

What was actually wrong

NOTES

Why the proposal was wrong, or what evidence confirmed it

Save decision

Recorded outcome

WHAT IT WASTwo hosts were statically configured with the same IP address 172.16.4.50.

FAULT TYPEOTHER

OSI LAYERLayer 3

8. Metrics dashboard

</dashboard>

The measurement screen: case counts, faults by type and by severity, the breakdown of human decisions, how often the AI agreed with the reviewer, and the responsible-AI log of every correction a human has made.

